

School: Computer Science
Institution: University of Windsor
Term: Winter 2026
Course: Comp-3150-1 : Database Management Systems
Instructor: Dr. C. I. Ezeife
Assignment #1 Solution: Total: 50 marks
Handed Out: Thurs. Jan. 15, 2026; **Due:** Thurs. Jan. 29, 2026

Objective of Assignment: To test on knowledge of database concepts and its 3-level architecture necessary for designing databases and their applications as well as practice on use of entity-relationship (ER) model to design databases.

Scope: Assignment covers materials from Chapters 1, 2 and 3 of book discussed in class.

Electronic Assignment Submission: Done through <http://brightspace.uwindsor.ca>

Marking Scheme : The mark for each of the questions is indicated beside each question.

Academic Integrity Statement: Remember to submit only work that is yours and include the following confidentiality agreement and statement at the beginning of your assignment.

CONFIDENTIALITY AGREEMENT & STATEMENT OF HONESTY

I confirm that I will keep the content of this assignment/examination confidential.

I confirm that I have not received any unauthorized assistance in preparing for or doing this assignment/examination. I confirm knowing that a mark of 0 may be assigned for copied work.

Student Signature

Student Name (please print)

Student I.D. Number

Date

Marking Scheme : The mark for each question and sub question is shown with the question below. Place your solutions in tables provided for answers where possible.

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Question	Mark
1	/10
2	/10
3	/10
4	/20
Total	/50

CHAPTER 1: DATABASES AND DATABASE USERS

- Given the simple Farmer-growscrops-inRegions database schema, which contains three files/tables as described below, answer the following questions with regards to this database.
(Total for que 1 is 10 marks)

Farmer (Ssn: integer, Name: string, Age: integer, Regionid: integer)

Growscrops (Ssn: integer, Regionid: integer, Crop: string, Revenue: real)

Region (Regionid: integer, Regname: string, Cropbudget: real, managerid: integer)

Note : Ssn, Name, Age, Regionid are the social security number, name, age and Regionid of the farmer respectively. Also, Regname, Cropbudget and managerid represent the regionid (e.g., SWO for south western Ontario), all crop budget (amount) for growing in this region (e.g., 100 for \$100M), and managerid (a farmer who oversees farmer affairs in this region). Revenue from grown crop can be measured in millions of dollars as well (e.g., 0.5 means half a million). A farmer may grow crop in multiple regions.

- Create a valid instance of this database containing values for its records with at least four records in each file.
(3 marks)
- Provide 2 informal English (not SQL) queries from this database with their answers. Each query should involve/use at least 2 of the files in the database and your answer should indicate the files (e.g., Farmer, Grows) needed to answer each query and specify what fields (attributes) are being retrieved as the result (e.g., Name, Age). Please, provide your solution in the 3 column table below.
(4 marks)
- Specify at least 3 relationships (one for each of the 3 database files) among the records of the database. That is, for each file (e.g., Farmer), list any relationships it has with other files through its fields (attributes) (e.g., Ssn). Provide your solution using the table below.
(3 marks)

Solution : (10 marks for que 1)

Query	Answer	Files involved																																
1. Create a valid instance of this database containing values for its records with at least four records in each file. (3 marks)	An instance of the Farmer-growscrops-inRegions database is : <u>Farmer</u> <table><tr><th><u>Ssn</u></th><th>Name</th><th>Age</th><th>Regionid</th></tr><tr><td>10</td><td>Jobe Bata</td><td>65</td><td>1</td></tr><tr><td>20</td><td>Monica Kap</td><td>80</td><td>1</td></tr><tr><td>30</td><td>Peter Good</td><td>22</td><td>2</td></tr><tr><td>40</td><td>Kate Lee</td><td>47</td><td>3</td></tr><tr><td>50</td><td>Ted Tam</td><td>50</td><td>4</td></tr></table> <u>Growscrops</u> <table><tr><th><u>Ssn</u></th><th><u>Regionid</u></th><th><u>crop</u></th><th>Revenue</th></tr><tr><td>10</td><td>1</td><td>corn</td><td>2.50</td></tr></table>	<u>Ssn</u>	Name	Age	Regionid	10	Jobe Bata	65	1	20	Monica Kap	80	1	30	Peter Good	22	2	40	Kate Lee	47	3	50	Ted Tam	50	4	<u>Ssn</u>	<u>Regionid</u>	<u>crop</u>	Revenue	10	1	corn	2.50	Farmer Growscrops Region
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	<table><tr><td>10</td><td>1</td><td>carrots</td><td>4.00</td></tr><tr><td>20</td><td>1</td><td>potatoes</td><td>9.00</td></tr><tr><td>30</td><td>2</td><td>potatoes</td><td>11.00</td></tr><tr><td>40</td><td>3</td><td>peas</td><td>15.30</td></tr><tr><td>50</td><td>4</td><td>potatoes</td><td>10.30</td></tr><tr><td>50</td><td>3</td><td>peas</td><td>14.00</td></tr></table> <table><tr><th colspan="4"><u>Region</u></th></tr><tr><th><u>Regionid</u></th><th>Regname</th><th>Cropbudget</th><th>managerid</th></tr><tr><td>1</td><td>SWO</td><td>5.00</td><td>10</td></tr><tr><td>2</td><td>EO</td><td>4.00</td><td>10</td></tr><tr><td>3</td><td>SC</td><td>9.00</td><td>50</td></tr><tr><td>4</td><td>SEO</td><td>6.00</td><td>50</td></tr></table>	10	1	carrots	4.00	20	1	potatoes	9.00	30	2	potatoes	11.00	40	3	peas	15.30	50	4	potatoes	10.30	50	3	peas	14.00	<u>Region</u>				<u>Regionid</u>	Regname	Cropbudget	managerid	1	SWO	5.00	10	2	EO	4.00	10	3	SC	9.00	50	4	SEO	6.00	50	
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2. Provide 2 informal English queries (not SQL) from this database with their answers. Each query should involve at least 2 of the files in the database and your answer should indicate the files (e.g., Farmer, Growscrops) needed to answer each query and specify what fields are being retrieved as the result (e.g., Name, Age). (4 marks) i. List the names, region names and city of farmers who have potatoes crop (Name, Regname, city). ii. Get the managerid, name of each manager and the total	<table><tr><td colspan="4">Result of query</td></tr><tr><td colspan="4">i.</td></tr><tr><td>Name</td><td colspan="3">Regname</td></tr><tr><td>Monica Kap</td><td colspan="3">SWO</td></tr><tr><td>Peter Good</td><td colspan="3">EO</td></tr><tr><td>Ted Tam</td><td colspan="3">SEO</td></tr><tr><td colspan="4">ii.</td></tr><tr><td>managerid</td><td>Name</td><td colspan="2">sum(Cropbudget)</td></tr><tr><td>10</td><td>Jobe Bata</td><td colspan="2">9.000.000</td></tr></table>	Result of query				i.				Name	Regname			Monica Kap	SWO			Peter Good	EO			Ted Tam	SEO			ii.				managerid	Name	sum(Cropbudget)		10	Jobe Bata	9.000.000		(i) Farmer Growscrops Region ii Farmer Region												
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3. Specify at least 3 relationships (one for each of the 3 database files) among the records of the database. For each file (e.g., Farmer), list any relationships it has with other files through its fields (e.g., Ssn). (3 marks)	Each Farmer record is related to one or more Growscrops records through the fields Ssn. Each Growscrops record is related to one Farmer record through Ssn and one Region record through Regionid. Each Region record is related to several Growscrops records through the field Regionid. Each Region record is also related to a Farmer record through its managerid attribute.	Farmer Growscrops Farmer Growscrops Region Farmer Growscrops Region

2. Recall that a database has many types of users, each of whom may require a different view of the database. For example, one user of the Farmer-growscrops-inRegions database of question 1 may be accessing and printing the details and Regionid of each Farmer frequently and thus a view for this user is created. Another view for this database is checking that each region has available budget on farming equipment and resources before expenditure such as for buying such equipment like farm tractors, seeds, etc.

(Total for que 2 is 10 marks)

- i) Using this Farmer-growscrops-inRegions database, give 2 additional views that may be needed by other user groups for the database.

(5 marks)

Solution : (5 marks for que 2i)

- (a) A view that groups all the farmers who grow crops in each region
- (b) A view that gives the total number of farmers who have generated total revenues of more than \$10million from their crops.

- ii) Give 5 examples from many of the different types of key, domain, foreign key, entity and semantic integrity constraints that you think can apply to the Farmer-growscrops-inRegions database of question 1.

(5 marks)

Solution : (5 marks for que 2ii)

- GA : any 5 of the following can be used.
- (a) The Ssn should be unique for each Farmer record (key constraint).
- (b) The Regionid should be unique for each Region record (key constraint).

- (c) A value of Ssn in a Growscrops record must also exist in Farmer record (referential integrity constraint).
- (d) A value of Regionid in a Growscrops record must also exist in Region record (referential integrity constraint).
- (e) The value of Regionid in a Growscrop record must be one of the values in the set {1, 2, 3, 4} (domain constraint).
- (f) Every record in Farmer must have a value for Ssn (entity integrity constraint).
- (g) A person must have age greater or equal to 18 to be seen as an adult farmer (general semantic integrity constraint).

CHAPTER 2: DATABASE SYSTEM CONCEPTS AND ARCHITECTURE

3.a. Design a simple database schema with 4 or less files for a University database system indicating all applicable constraints and information. In this University, students have majors and take courses which they receive grades for. These grades are used to compute the student grade point average at any point in time. As the database designer, you should decide the necessary attributes for students and courses. Also, show a sample database state for the database.

(5 marks for a)

b. Using your database, describe the differences between logical and physical data independence.

(5 marks for b)

(Total for que 3 is 10 marks)

Question	Answers																														
a. Design a simple database schema with 4 or less files for a University database system indicating all applicable constraints and information. Also, show a sample database state for the database. (5 marks)	'Students take courses' database schema is : Student (stuid : integer, sname : string, major : string, gpa : real) Take (stuid : integer, cid : string, grade : integer) Course (cid : string, ctitle : string) Some constraints are : A student can take many courses. A course can be taken by many students. A state of this database is : Student <table><tr><th>Stuid</th><th>sname</th><th>major</th><th>gpa</th></tr><tr><td>11</td><td>John Smith</td><td>CS</td><td>80</td></tr><tr><td>22</td><td>Mary Cane</td><td>Math</td><td>67</td></tr></table> Take <table><tr><th>Stuid</th><th>cid</th><th>grade</th></tr><tr><td>11</td><td>60-140</td><td>60</td></tr><tr><td>11</td><td>60-100</td><td>70</td></tr><tr><td>11</td><td>62-120</td><td>75</td></tr><tr><td>22</td><td>62-140</td><td>80</td></tr><tr><td>22</td><td>40-140</td><td>90</td></tr></table>	Stuid	sname	major	gpa	11	John Smith	CS	80	22	Mary Cane	Math	67	Stuid	cid	grade	11	60-140	60	11	60-100	70	11	62-120	75	22	62-140	80	22	40-140	90
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b. Using your database, describe the differences between logical and physical data independence	Logical data independence: i. The ability to change the conceptual schema (e.g. get names of students with GPA>85%; and get names and address of CS students) without having to change the external schema or application program (eg. when student has an additional attribute "address"). ii. Physical data independence: It has the ability to change the internal schema (e.g., storage model like store the files as B-tree instead of arrays) without having to change the conceptual schema such as relations. For example an access path (such as B-tree) to improve retrieval speed of TAKE file records should not require the TAKE relation or file or its query to be altered much. An example query on TAKE is "list all courses taken by each student".												

CHAPTER 3: DATA MODELING USING THE ENTITY-RELATIONSHIP (ER) MODEL

4 You have been hired to design a database for another version of the Farmer-growscrops-inRegion database world and your first job now is to design an ER model for this database using the following description of that world.

- i. farmers are identified by their SSN and have other attributes as names, ages and Regionid.
- ii. regions are identified by their region id and have other attributes as region names, cropbudget and manager id.
- ii. Each crop is identified by the crop id, name and crop type.
- iii. Each farmer can grow several crops in more than one region and each crop can be grown by many farmers
- iv. Each region can have several farmers growing crops in it.
- v. Each region can have several crops growing in it.

Design the Entity-Relationship (ER) model diagram for this database.

(Total for que 4 is 20 marks)

(Note : 10 marks for correct entity and relationship identifications with their attributes in ER (5 for entities with attributes and 5 for relationships with attributes), 5 marks for correct cardinality/participation constraints interpretations on the edge labels, 5 marks for correct verbal interpretations of the database being represented by the ER digram through use of correct

symbols etc.). If all components above are presented with NO ER DIAGRAM, LOSE 15 MARKS. If only the correct ER diagram is presented with no conceptual design information about it as given in the table below, lose 10 marks.

Hint : Present the conceptual design first, showing (1) all the entities and their attributes, (2) all the relationships and their attributes, (3) all the constraints before drawing your ER.

(Total for que 4 is 20 marks) Conceptual information in table and ER diagram next

Specific Requirement/Constraint Type	Requirements and Constraints from the ER diagram
Entities and attributes (5 marks) in ER	Farmer (<u>Ssn</u> , Name, Age, Regionid) Crop(<u>Cropid</u> , Cname, type) Region (<u>Regionid</u> , Regname, Cropbudget, managerid)
Relationships and attributes (5 marks) in ER	Growscrops (<u>Ssn</u> , <u>Regionid</u> , <u>Crop</u> , Revenue)
Interpretation of each of the constraints represented on the edge labels (5 marks) in ER	1. Each farmer can grow several crops in more than one region and each crop can be grown by many farmers. 2. Each region can have several farmers growing crops in it. 3. Each region can have several crops growing in it.
Correct use of symbols in ER, etc (5 marks)	Show correct use of symbols for attributes, relationships, participations, etc.

ER Diagram goes next :

You may attach a scanned copy of your hand-drawn ER diagram here. You can also draw it digitally if possible, and attach. Note that in the ER diagram, the foreign key attributes that are part of the relationship schemas are not explicitly listed with the relationship but inherited from the entity the relationships are connected to.

The ER model Diagram for the Farmer-growscrops Database of Question 3 of Assignment 1 is:.

